Batuhan Altundas

470-313-0313 | baltundas3@gatech.edu
linkedin.com/in/altundasbatu | github.com/batuhanaltundas

Summary

Computer Science Ph.D. student researching machine learning for robotics and automation. Develops graph neural networks and reinforcement learning methods for multi-agent task allocation and scheduling. First-author publications at NeurIPS and IROS; seeking a research internship.

Education

Georgia Institute of Technology | Atlanta, GA | Aug 2023 - Present
Doctor of Philosophy (Ph.D.), Computer Science

Georgia Institute of Technology | Atlanta, GA | Aug 2020 - May 2022
M.S., Computer Science (Computational Perception and Robotics)

University of Michigan | Ann Arbor, MI | Aug 2014 - Dec 2019
B.S.E., Computer Science and Mechanical Engineering (Dual Major)

Technical Skills

Programming: Python, C++, SQL, MATLAB
ML Frameworks: PyTorch, TensorFlow, Deep Graph Library (DGL), Pyro
Machine Learning: Deep learning, reinforcement learning (RL), graph neural networks (GNNs), transformers, imitation learning, behavior cloning, large language models (LLMs)
Robotics: Robot Operating System (ROS), Gazebo, OpenAI Gym, simultaneous localization and mapping (SLAM), computer vision, motion planning, autonomous navigation
Tools: Linux, Git, Docker, Azure Databricks, Weights & Biases, Tensorflow, Raspberry Pi

Research Experience

Graduate Research Assistant
Georgia Institute of Technology, CORE Robotics Lab | Atlanta, GA | Aug 2023 - Present

- Design and train graph neural networks, heterogeneous graph transformers, and recurrent reinforcement learning policies for multi-agent task allocation, scheduling, and path planning.
- Developed TARGETNET for simultaneous assignment and scheduling, evaluated on up to **40 agents and 200 tasks**, with **20-250x speedups** over prior graph-based methods across benchmark scales (NeurIPS 2025).
- Derived interpretable decision tree representations of learned task allocation and scheduling policies.

Research Scientist I
Georgia Institute of Technology | Atlanta, GA | Jul 2022 - Aug 2023

- Led graduate researchers in multi-agent task allocation, scheduling, and motion planning.
- Developed simultaneous scheduling methods that replace task-by-task model evaluations with a single forward pass for joint assignment and ordering.

Industry Experience

Data Science Intern
Halliburton | Houston, TX | Jun 2021 - Aug 2021

- Applied unsupervised learning and dimensionality reduction to industrial time-series data using Azure Databricks.
- Trained variational autoencoders (VAEs) and autoregressive neural networks for feature prediction and anomaly characterization.
- Presented modeling approaches and technical findings to nontechnical stakeholders.

Publications

- **Altundas, B.**, Chen, S., Singh, S., Deo, S., Cho, M., & Gombolay, M. Heterogeneous Graph Transformers for Simultaneous Mobile Multi-Robot Task Allocation and Scheduling under Temporal Constraints. *Conference on Neural Information Processing Systems (NeurIPS)*, 2025.
- Natarajan, M., Seraj, E., **Altundas, B.**, et al. Human-Robot Teaming: Grand Challenges. *Current Robotics Reports*, 2023.
- **Altundas, B.**, Wang, Z., Bishop, J., & Gombolay, M. Learning Coordination Policies over Heterogeneous Graphs for Human-Robot Teams via Recurrent Neural Schedule Propagation. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022.

Research Projects

LLM-Based Multi-Agent Coordination in a Tabletop Game Environment
Python, LLaMA, GPT, reinforcement learning

- Built an adversarial multi-agent environment supporting both reinforcement learning policies and natural-language commands.
- Fine-tune foundation models and compare their performance with reinforcement learning agents in multi-agent settings.

Behavior Cloning for Fetch Robot Manipulation
Python, OpenAI Gym, imitation learning, learning from demonstration

- Developed a custom simulation environment for Fetch robot pick-and-place tasks with execution noise.
- Evaluated noise-free pretraining for behavior cloning before training policies in noisy environments.

M-Bot Autonomous Navigation
Python, computer vision, SLAM, Raspberry Pi

- Integrated line detection and computer vision with localized SLAM to model indoor corridors and correct robot heading.
- Developed vision-based intersection detection and integrated path finding for semi-autonomous indoor navigation.

Additional Experience

Graduate Teaching Assistant
Georgia Institute of Technology | Atlanta, GA | Aug 2021 - May 2022

- Supported instruction in CS 6476: Computer Vision for the Online Master of Science in Computer Science (OMSCS) program.

Research Assistant
University of Michigan | Ann Arbor, MI | Jun 2018 - Sep 2018

- Analyzed, modeled, and prototyped a robotic joint using a gearless clutch system for legged robots.

Patents

Altundas, B., Wang, Z., & Gombolay, M. Learning Coordination Policies over Heterogeneous Graphs for Human-Robot Teams via Recurrent Neural Schedule Propagation. U.S. patent application No. 63/592,374, 2023.